

# Requirement Analysis

# Objectives

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- A thick blue arrow pointing to the right, starting from the left edge of the slide and ending just before the first objective bullet point.
- Understand how to create a requirements definition.
  - Become familiar with requirements analysis techniques.
  - Understand when to use each requirements analysis technique.
  - Understand what and where to use REQUIREMENTS ANALYSIS STRATEGIES

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**Segment 1:** Requirements Specification

**Segment 2:** Requirement Gathering Techniques and Interview

**Segment 3:** JOINT APPLICATION DESIGN (JAD)

**Segment 4:** Other Requirement Gathering Techniques

# Key Ideas

- The goal of the analysis phase is to truly understand the requirements of the new system and develop a system that addresses them.
- The first challenge is **collecting and integrating the information**
- The second challenge is finding the **right people** to participate.

# Analysis Phase

- This phase takes the general ideas in the system request and
  - refines them into a detailed requirements definition (this chapter),
  - functional models
  - structural models and
  - behavioral models
- This becomes the system proposal
  - Includes revised project management deliverables,
    - feasibility analysis and
    - workplan

# Requirement Specification

- A ***requirement*** is a statement of what the system must do or what characteristics it needs to have
- Types of requirements:
  - What the business needs (business requirements)
  - What the users need to do (user requirements)
  - What the software should do (functional requirements)
  - Characteristics the system should have (nonfunctional requirements)
  - How the system should be built (system requirements)

# Functional Requirements

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- A process the system should perform as a part of supporting a user task, or
- Information the system should provide as the user performs a task
- Specify the support the system will provide to the user in fulfilling his/her work tasks

# Non-Functional Requirements

- Nonfunctional requirements are “the quality attributes, design, and implementation constraints, and external interfaces which a product must have”.
- This requirement category includes important behavioral properties that the system must have
- Behavioral properties the system must have
  - Operational – physical and technical operating environment
  - Performance – speed, capacity, and reliability needs
  - Security – access restrictions, needed safeguards
  - Cultural and political – issues that will affect the final system



# Functional Requirements example

## Functional Requirements

### 1. New Vehicle Management

- 1.1 The system will allow managers to view the current new vehicle inventory.
- 1.2 The system will allow the new vehicle manager to place orders for new vehicles.
- 1.3 The system will record the addition of new vehicles to inventory when they are received from the manufacturers.

### 2. Vehicle Sales Management

- 2.1 The system will enable salespersons to create a customer offer.
- 2.2 The system will allow salespeople to know whether an offer is pending on a specific vehicle.
- 2.3 The system will enable managers to record approval of a customer offer.
- 2.4 The system will prepare a sales contract.
- 2.5 The system will prepare a shop work order based on customer requested dealer options.
- 2.6 The system will record a customer deposit.
- 2.7 The system will record a customer payment.
- 2.8 The system will create a record of the customer's vehicle purchase.

### 3. Used Vehicle Management

- 3.1 The system will record information on a customer trade-in vehicle ... etc.

# Nonfunctional Requirements example



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## Nonfunctional Requirements

### 1. Operational

- 1.1 The system should run on tablet PCs to be used by salespeople.
- 1.2 The system should interface with the shop management system.
- 1.3 The system should connect to printers wirelessly.

### 2. Performance

- 2.1 The system should support a sales staff of 15 salespeople.
- 2.2 The system should be updated with pending offers on vehicles every 15 minutes.

### 3. Security

- 3.1 No salesperson can access any other salesperson's customer contacts.
- 3.2 Only the owner and sales manager may approve customer offers.
- 3.3 Use of each tablet PC should be restricted to the salesperson to whom it is assigned.

### 4. Cultural and Political

- 4.1 Company policy says that all computer equipment is purchased from Dell.
- 4.2 Customer personal information is protected in compliance with the Data Protection Act.
- 4.3 The system will conform to the state's "lemon law."

# Types of Nonfunctional Requirements



| Requirement            | Description                                                                  | Examples                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational            | The physical and technical environments in which the system will operate     | <ul style="list-style-type: none"><li>• The system will run on Android mobile devices.</li><li>• The system should be able to integrate with the existing inventory system.</li><li>• The system should be compatible with any Web browser.</li></ul>                                                                                                                                                                   |
| Performance            | The speed, capacity, and reliability of the system                           | <ul style="list-style-type: none"><li>• Any interaction between the user and the system should not exceed 2 seconds.</li><li>• The system downloads new status parameters within 5 minutes of a change.</li><li>• The system should be available for use 24 hours per day, 365 days per year.</li><li>• The system supports 300 simultaneous users from 9–11 a.m.; 150 simultaneous users at all other times.</li></ul> |
| Security               | Who has authorized access to the system under what circumstances             | <ul style="list-style-type: none"><li>• Only direct managers can see staff personnel records.</li><li>• Technicians can see only their own work assignments.</li><li>• The system includes all available safeguards from viruses, worms, Trojan horses, etc.</li></ul>                                                                                                                                                  |
| Cultural and Political | Cultural and political factors and legal requirements that affect the system | <ul style="list-style-type: none"><li>• The system should be able to distinguish between US currency and currency from other nations.</li><li>• Company policy is to buy computers only from Dell.</li><li>• Country managers are permitted to authorize custom user interfaces within their units.</li><li>• Personal information is protected in compliance with the Data Protection Act.</li></ul>                   |



| Sl.No | Functional Requirement                                                                                                  | Non-functional Requirement                                                                                                                                                                    |
|-------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .1    | Defines all the services or functions required by the customer that must be provided by the system                      | .Defines system properties and constraints e.g<br>.reliability, response time and storage requirements<br>Constraints are I/O device capability, system<br>.representations, etc              |
| .2    | .It describes what the software should do                                                                               | It does not describe what the software will do, but<br>.how the software will do it                                                                                                           |
| .3    | Related to business. For example: Calculation of order value by Sales Department or gross pay by the Payroll Department | .Related to improving the performance of the business<br>For example: checking the level of security. An operator should be allowed to view only my name and<br>.personal identification code |
| .4    | .Functional requirement are easy to test                                                                                | Nonfunctional requirements are difficult to test                                                                                                                                              |
| .5    | Related to the individual system features                                                                               | Related to the system as a whole                                                                                                                                                              |
| .6    | Failure to meet the individual functional requirement may degrade the system                                            | Failure to meet a non-functional requirement may<br>.make the whole system unusable                                                                                                           |

## Segment 2

### REQUIREMENTS ANALYSIS STRATEGIES

# Requirement Gathering Techniques

- An analyst search for requirements using a variety of techniques
- Make sure that the current business processes and the needs for the new system are well understood before moving into design.
- Five most commonly used requirements elicitation techniques:
  1. Interviews
  2. JAD sessions
  3. Questionnaires
  4. Document analysis
  5. Observation.

# REQUIREMENTS ANALYSIS STRATEGIES

# Problem Analysis

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- Ask users to identify problems and solutions
  - Improvements tend to be small and incremental
  - Rarely finds improvements with significant business value



# Root Cause Analysis

- Users are not asked for solutions, but for:
  - A list of (prioritized) problems.
  - All possible root causes for those problems.
- Analysts investigate each root cause to find:
  - Solutions for the highest priority problems.
  - Root causes that are common to multiple problems.

# Duration Analysis

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- Calculate time needed for each process step
- Calculate time needed for overall process
- Compare the two – a large difference indicates a badly fragmented process
- Potential solutions:
  - Process integration – change the process to use fewer people, each with broader responsibilities
  - Parallelization – change the process so that individual step are performed simultaneously

# Activity-Based Costing

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- Calculate cost of each process step
- Consider both direct and indirect costs
- Identify most costly steps and focus improvement efforts on them

# Benchmarking

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- A thick blue arrow pointing to the right, spanning the width of the slide, positioned above the list of bullet points.
- Studying how other organizations perform the same business process
  - Informal benchmarking
  - Common for customer-facing processes
  - Interact with other business' processes as if you are a customer

# Technology Analysis

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- Analysts list important and interesting technologies
- Managers list important and interesting technologies
- The group identifies how each might be applied to the business and how the business might benefit

# Activity Elimination

- Identify what would happen if each organizational activity were eliminated
- Use “force-fit” to test all possibilities

# Comparing Analysis Techniques

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- Potential business value
- Project cost
- Breadth of analysis
- Risk



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# Thank you